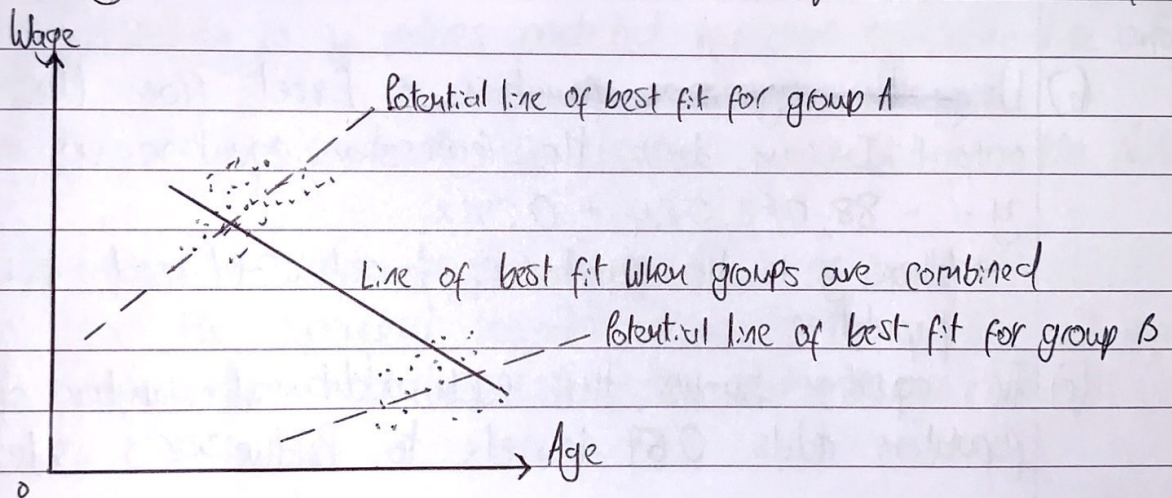


Supervision 3

Q1. $r = 0.8$, using the CORREL(...) function in Excel

Q2. Using Excel, I found the correlation to be $r = -0.43$. This result seems strange, as one would expect a positive correlation between age and wage owing to the potential for skills development and promotions over time. Plotting a scatter graph in Excel I noticed the data set seems to be composed of two distinct data points that on their own exhibit a positive correlation, but combined show a negative one. This suggests these groups have different occupations, and should therefore be analysed separately to draw out the true relationship between X and Y.



Q3. → Strong correlation between time at company and Monthly Sales indicated by a correlation of 0.92. Regression equations means for each extra month at the company brings expected increase in Sales of 0.55. However our most extreme data point is for 50 months, so extrapolating to 72 months is not advised as there is no way to know if the relationship will hold for this value.

- The $r = -0.03$ correlation between Monthly and Mileage should spark suspicion as a manager would expect employees who are travelling more to make more sales.
- We can quickly identify Adams as anomalous to the other representatives (data points). He has highest mileage and lowest sales. Without him the correlation becomes 0.944 which is highly significant.

Q4.

(a) Scatter plot developed in Excel

$$y = 0.6911x - 9E+07$$

$$R^2 = 0.9349$$

(b) Using the regression function in Excel, from the summary output I can derive the regression equation as

$$y = -88,062,524 + 0.691x$$

→ Where y is the number of barrels sold and x is the US population

(c) This equation shows that each additional member of the US population adds 0.69 barrels to Budweiser's sales

Q5. I believe a country's average daily temperature and the average heating bill of its residents. This is because in countries with higher temperatures residents will likely spend a smaller proportion of time needing to heat their homes. Mathematically, this means the correlation coefficient between average temperature and heating bill is $-1 < r < 0$.

Q6. The correlation coefficient shows the strength of the linear relationship between the two variables. If r is close to 0, the variables are not strongly correlated, and

if the r absolute value is near 1, the variables are strongly correlated.

The R-square statistic is the proportion of variation in the dependent variable that can be explained by the ~~independent~~ ~~variable~~ regression equation. ~~It~~ In other words, it shows us to what extent variance in y is explained by variance in x . Its equation is $R^2 = 1 - \frac{\text{Unexplained Variation}}{\text{Total Variation}}$

$$\text{or } R^2 = \frac{1 - (\text{Total variation} - \text{Explained Variation})}{\text{Total Variation}}$$

Total variation is the residual between data points and y bar
Explained variation is y values predicted by our regression line and y bar

→ Closer R^2 is to 1, the better our model fits the data

The slope of a simple regression equation, or line of best fit, shows how the dependent variable responds to a unit change in the independent variable within the data range analysed

Q7.

(a) Yes. The summary output shows a Multiple R , or correlation coefficient of 0.75. This is high, especially given sample size, thus indicates a positive relationship.

(b) Regression equation: $Y = 853.2918 + 627.6459X$

$$853.2918 + 627.6459(52) = £33490.88$$

Estimation of 95% confidence interval: $[Y - 2SE, Y + 2SE]$

$$SE = 17735$$

95% confidence interval for the operation costs of an outlet selling 52 TVs is $-1980 \leftrightarrow 68961$, or 0 to £68,961 as you can't have negative costs

(c) $Y = 853.3 + 627.6X$

$Y = 44785.3 \rightarrow$ Expected y value

$[Y - 2SE, Y + 2SE] = 9,315 \leftrightarrow 80,256$

€90,000 is outside this range so with 95% confidence level we conclude this outlet's costs are significantly higher than the normal range

Multiple regression Questions

Q1.

(a) $Y = 29.05 + 34.507i - 1.691k - 0.082a$

(b) The adjusted R -square shows us that 67% of the variance in the amount of life insurance held is explained by variance in income, number of kids and age

(c) A p -value of < 0.05 shows us that the explanatory variable has a significant effect on the dependent variable, thus we can say incomes positively effect insurance held.

"kids" and "age" coefficients have p -values > 0.05 , so do not appear to have a statistically significant effect on insurance held. That said, there may be collinearity between the variables, so we could remove one from the model

Q2.

- (a) Done in Excel
- (b) Estimated regression equation: $Y = 3750.5 + 0.0995x$
- (c) $Y = 3750.5 + 0.0995(2,500) = 3999.25 \approx 3999$
- (d) $Y = ~~4158~~ 4099$
- (e) For each one dollar increase in budget, output per worker would increase by 0.0995 according to the regression equation
- (f) Using the summary output, the 95% confidence interval for the slope is $[0.04578, 0.153304]$

Q3.

- (a) Population and DVD library are significant predictors of Sales as they have p-values of ≤ 0.05 .
- (b) Looking at the coefficient, an additional DVD increases sales by 1.35 per month
- (c) $[0.26, 2.44]$
- (d)